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House sparrow (*Passer domesticus*) habitat use in urbanized landscapes

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Abstract The house sparrow (*Passer domesticus*) is showing population declines in many parts of Europe, with recent declines being particularly severe in urban areas. To date, relatively little is known about the species' habitat associations within urbanized landscapes. We report here an investigation of the habitat associations of house sparrows using a survey of 1223 stratified randomly selected 500 × 500-m squares within urbanized landscapes of the UK, defined as at least 25% 'human cover'. The densities of chirping male house sparrows and of all house sparrows were analysed separately to obtain insights into breeding habitat requirements and general habitat associations, respectively. Multi-model inference showed that residential areas (houses, flats), allotments (areas used for small-scale horticulture) and farm buildings were key predictors of house sparrow density and chirping male density. Separate analyses on landscapes of differing human cover showed similar results. Within residential areas, the increase of house sparrow density with habitat area (on a log scale) was approximately threefold greater when private gardens were present than when they were absent. The model predicted a rapid decline in house sparrow abundance when only a small area of private gardens is converted to continuous housing. Allotments and residential areas with gardens are

likely to be under pressure due to increased demand for housing, specifically from the infilling of green space within urban areas. It would seem to be imperative that any action plan to protect urban house sparrow populations should include specific protection of such key habitats.

Keywords Allotments · Gardens · Green space · Urbanization

Introduction

The house sparrow *Passer domesticus*, a species once so numerous that it was considered a pest in many parts of Europe, is now showing severe population declines in many areas (Crick et al. 2002; Summers-Smith 2003; Anderson 2006). In the UK, the species underwent an estimated 68% decline between 1977 and 2003 (Baillie et al. 2006), which has resulted in the species being placed on the Red List of species of conservation concern (Gregory et al. 2003). This estimate is, however, biased towards farmland within more heavily populated areas of the southern and eastern UK (Fuller et al. 1985). More recent analyses suggest that house sparrow population changes show marked regional variation and are particularly severe in urbanized environments (Siriwardena et al. 2002; Cannon et al. 2005; Chamberlain et al. 2005), although there appears to be much variation between populations in individual cities. For example, declines have been detected in London, Dublin, Glasgow, Edinburgh, Ghent (Summers-Smith 2003) and Hamburg (Mitschke 1999), but there have been no such declines in Berlin, Paris or Manchester (Summers-Smith 2003).

Population declines on farmland seem most likely to be influenced by decreases in survival due to changing

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farming practice affecting food availability (Siriwardena et al. 2002; Hole et al. 2002). The timing of declines in urban areas differs from those in farmland, and the mechanisms underlying the former are likely to be different (Siriwardena et al. 2002; Summers-Smith 2003). Although a number of hypotheses have been put forward to explain these declines, including increased predation, effects of pollution, reduction in food chick supply and reduced availability of nest sites (Summers-Smith 2003; Anderson 2006), there is as yet no strong evidence to support any of these. The results of some studies, however, may point towards the most likely causes. A pattern that appears in some city populations is that house sparrows are most abundant and population declines are lower in socially deprived areas (Dröscher 1992; Bland 1998; Paston 2000; Robinson et al. 2005). This could be due to several factors, including more waste ground and gardens that have less management (e.g. pesticide inputs), leading to greater food availability, and fewer home improvements leading to a greater availability of nest sites. Wooton et al. (2002) found that nesting house sparrows were more likely to occur in older properties (built before 1919) and in medium-aged properties (20–60 years old) that had not had roof repairs, suggesting a link to nest site availability. Wilkinson (2006) found the probability of house sparrow occurrence in a suburban area to be positively related to bush cover and, in particular, to native species of bush, which may imply effects of food availability. Similarly, Heij (1985) suggested that the amount of green space was important in determining house sparrow density in urbanized environments but that there was also a correlation with building density, showing the importance of feeding and nesting sites in close proximity to each other. However, Mason (2006) found no significant correlation between house sparrow density and either the area of urban green space (including gardens) or housing density.

In this paper we examine the habitat associations of house sparrows in predominantly urbanized landscapes using data from a large volunteer-based survey. The goal of the paper is to identify key habitats for house sparrows in urbanized environments. More generally, the data set will act as an invaluable baseline against which to compare future changes in house sparrow populations in relation to habitat change.

Methods

Survey design and stratification

Stratified random sampling techniques were employed in order to target sufficiently representative 1-km squares of urbanized habitat in the UK. Countryside (CS) 2000

landcover data (Fuller et al. 2002) were used to examine coverage by the habitat categories ‘suburban/rural development’ and ‘continuous urban development’, combined to form one class named ‘human cover’. This variable was used to stratify the data set for each of 266,000 1-km squares within the UK into classes of human coverage. Based on the cumulative square root $f(y)$ rule (Krebs 1989) and on visual inspection of square distribution, three strata of human cover were defined: A, 25–49.9% (capturing 9502 of all UK 1-km squares); B, 50–74.9% (capturing 5804 of all UK 1-km squares); C, 75–100% (capturing 4564 of all UK 1-km squares). Squares with <25% human cover were not selected owing to the extreme lack of urban habitat suitable for house sparrows.

Optimal allocation was used to determine the requisite number of squares for each of the three strata, based on house sparrow density data from a pilot study, assuming a constant sampling cost. To ensure proportional geographic representation, data sets containing all relevant squares for each stratum were ordered by British grid reference. The data were then partitioned into subsets, and one square was randomly selected from each sub-sample. Following this procedure, the number of squares allocated to each stratum was examined at a regional level, based on Government Office Regions. Random re-sampling occurred where necessary so that each region/stratum combination contained at least 44 targeted squares. In total, 2420 squares were targeted for coverage; 997 squares from stratum A, 762 from stratum B and 661 from stratum C.

Volunteer selection

To maximize take-up of survey squares, the Geographic Information Service (GIS) was used to identify the four nearest squares to the residential home square of each participant in the BTO/CJ Garden BirdWatch scheme (Toms 2003), with an upper limit of 5 km between the edges of residential and targeted grid squares. In this way, it was possible to match volunteers to proximate squares for house sparrow survey coverage and approach the most likely volunteers.

Bird recording

Observers were asked to walk along all pavements, paths and roads, into parks and allotments (small areas within or on the edges of urban settlements that are leased to the public for small-scale horticulture) and along field boundaries with the aim of mapping the location of all house sparrows detected (in the habitat patch in which they were first seen or heard). Chirping males, other males and females were each recorded separately. Individual sparrows for which the sex could not be determined were also

recorded. The recommended start time was within 2 h of sunrise. Two visits were carried out in the summer of 2003, one in May and one in June (with a minimum of 1 week separating the two visits), one in the autumn of 2003 (October) and one in the summer of 2004 (May or June).

Habitat data

Before the bird survey visits, surveyors were required to undertake a habitat recording visit. All roads and paths in the survey squares were walked and habitat was recorded into discrete patches that were over “half a tennis court in size” (approx. 130 m²). These patches were drawn as accurately as possible onto the site map, and each patch was given a number corresponding to a pre-selected list of 30 habitat types (Appendix A in Supplementary Material). This list was reduced to 13 habitat variables (e.g. by combining ecologically similar habitat types) which are listed in Table 1 (inclusion of all 30 habitat types, many of which were recorded at low frequencies, caused analytical problems). The habitat data were entered into a GIS, and patch areas for each habitat type were determined for each survey square. For the analysis, very small habitat patches (<1 ha in area) were excluded in order to avoid misleadingly high densities as a result of small patch size.

Analysis

In this paper, the densities of chirping male house sparrows and of all house sparrows were analysed separately to provide insights into breeding habitat requirements and general habitat associations, respectively. House sparrow count was modelled in relation to visit, habitat and land-

cover variables using PROC GENMOD in SAS (SAS Institute, Cary, N.C.). Initial analyses with Poisson regression indicated that the data were over-dispersed (i.e. the variance was not equal to the mean), as measured by deviance/degrees of freedom. The negative binomial distribution, with a log link function, was used as an alternative and provided a good model fit (deviance/df < 2) in all cases and was therefore used as the standard model.

Not all squares received four survey visits (see below). Initial analyses were performed to see if the squares that were visited less frequently had lower numbers of house sparrows. This was analysed by determining the number of house sparrows at the whole square level detected on Visit 1 (‘first count’) and seeing if this differed between squares with differing numbers of visits.

Habitat associations were first analysed at the habitat patch level, where the count in each patch was related to the habitat in that patch. Patch area (log-transformed) was used as an offset in the models. As there were usually several patches per survey square, the latter was defined as a subject variable in a repeated measures model, and parameter estimates and significance tests were derived from general estimating equations (GEE). This ensured that the parameter estimate confidence limits were adjusted to account for the correlation between observations from the same square that could have been caused by unmeasured factors (e.g. individual observer) associated with the square.

A second approach was taken whereby the area of each habitat type was analysed as a separate continuous variable in relation to house sparrow count at the whole square level. The key predictor variables were identified using multi-model inference (Burnham and Anderson 2002),

Table 1 Habitat types recorded in the house sparrow survey and the mean area of each habitat per patch

Habitat code ^a	Definition	Notes	Mean ± SE patch area (ha)	N ^b
ALLOT	Allotments		1.88 ± 0.12	275
ARABL	Arable farmland		6.73 ± 0.25	660
BROWN	Brownfield sites	Includes ruined buildings	1.06 ± 0.06	919
BUILD	Buildings (commercial/industrial)	For example, shops, factories, bus and train stations, petrol stations	1.97 ± 0.06	2,744
FARMS	Farm buildings		0.54 ± 0.04	230
HOUSE	Residential area with gardens		7.98 ± 0.04	18,366
FLATS	Residential area without gardens		2.22 ± 0.10	767
GRASL	Grass with livestock		3.50 ± 0.18	603
GRASS	Grass without livestock		3.69 ± 0.16	828
LINER	Linear byways	Roads, paths, rail lines	1.04 ± 0.02	2,087
OCOUN	Miscellaneous open countryside		3.39 ± 0.22	493
PARKS	Parks and water bodies		1.78 ± 0.05	2,849
WOODS	Woodland		1.59 ± 0.07	1,655

^a Each habitat is given a five-letter code used in subsequent figures and tables

^b N is the total number of habitat patches out of 1223 survey squares

whereby models of all possible combinations of habitat variables are compared using the weighted Aikake information criterion (AIC) value. The summed model weights (Σwi) across all models give a measure of the importance of each variable. Σwi has a value between 0 and 1, where higher values indicate variables that were more likely to be included in the ‘better’ models as measured by the weighted AIC value. The size of effect of each variable was determined using model averaging (Burnham and Anderson 2002), where the mean and 95% confidence limits of parameter estimates across all models for a given variable are determined. Consistent effects of a given variable across models are demonstrated when confidence limits do not overlap 0.

The above procedure was repeated separately for survey squares classified as either category A (low-density human cover) and category C (high-density human cover) in order to determine whether house sparrow habitat associations varied across a gradient of human land-use intensity within a generally urbanized landscape. The scale of the house sparrow survey was 0.25-km², so it is assumed that land-cover at 1 km² is representative of landcover at 0.25 km².

Results

Survey bias

A total of 1223 squares were surveyed. Four visits were undertaken, but not every square received four visits. The number of squares surveyed for Visits 1–4 respectively was 1223, 1175, 918 and 736. There was a highly significant overall difference in the first count according to the number of survey visits a square received (Table 2), where squares that received only one visit had a much lower count than squares that were visited more often. These results suggest that observers were less likely to make repeat visits to a square if house sparrows were initially absent or scarce. For this reason, all statistical analyses are undertaken separately by visit, and the focus is on the first visit as that

is less biased towards habitats where house sparrows were numerous.

House sparrow density

The mean density of all house sparrows recorded in 13 habitat types recorded in the survey is shown in Fig. 1 for each visit. There were highly significant effects of habitat on house sparrow count in each visit ($p < 0.0001$). The highest mean densities were in allotments, residential areas—both with (HOUSE) and without (FLATS) gardens—and farm buildings, although the counts were very variable (large error bars) in the latter habitat type (Fig. 1). The major difference was in the autumn visit, where there were relatively lower counts in farmland. A further analysis was carried out comparing only the habitats with a sparrow density of $>1/\text{ha}$ (ALLOT, FARMS, FLATS, HOUSE). There were highly significant differences between habitats in these reduced models ($P < 0.0001$), with FLATS having the lowest estimated density in each visit (Table 3).

The mean count of chirping males in different habitats in each visit is shown in Fig. 2. Chirping males were very scarce in the autumn, and no analyses were carried out in this period. This is perhaps unsurprising given that this behaviour is primarily associated with breeding, although there can be a minor resurgence of sexual display in the autumn (Summers-Smith 1963). For the breeding season visits (Visits 1, 2 and 4), there were highly significant differences in each case ($P < 0.0001$), with highest counts in the farmland and residential areas, although allotments did not have such high relative densities compared with total house sparrows (Fig. 1), possibly suggesting they are used more for feeding than breeding. A repeat analysis on the four habitats with the highest density supported this, with allotments having the lowest estimated density in each of the breeding season surveys (Visits 1, 2 and 4; Table 3).

Multi-model averaging

Habitat associations of house sparrows at the whole square level were analysed using multi-model inference, considering only Visit 1 (i.e. the visit that was considered to be the least biased towards squares with higher numbers of house sparrows). There were no single models that stood out as having markedly higher model weights (wi). Rather, the ‘best’ models tended to have similar wi and typically had several predictor variables; therefore, it was difficult to identify a single ‘best’ model. For example, for all house sparrows, wi ranged from 0.05 to 0.08 for the ten highest ranked models, and these consisted of between six and eight predictor variables. We therefore focus on the model-averaged parameter estimates and Σwi , rather than on individual models.

Table 2 The house sparrow count on the first visit per survey square in relation to the number of visits to each square

Number of visits	First count ^a	Model details ^b
1	14.98 (12.90–17.41)	$\chi^2_4 = 46,517$
2	24.00 (22.67–25.41)	$p < 0.0001$
3	23.41 (22.18–24.70)	$D = 1.19$
4	24.05 (23.08–25.05)	

^a Estimates (95% confidence limits) are derived from generalized linear models with a binomial error structure

^b D is the deviance/ df

Fig. 1 Density of total house sparrows in different habitat types in different years/seasons. Error bars represent 95% confidence limits (note that in some cases upper confidence limits are not shown due to excessive size)

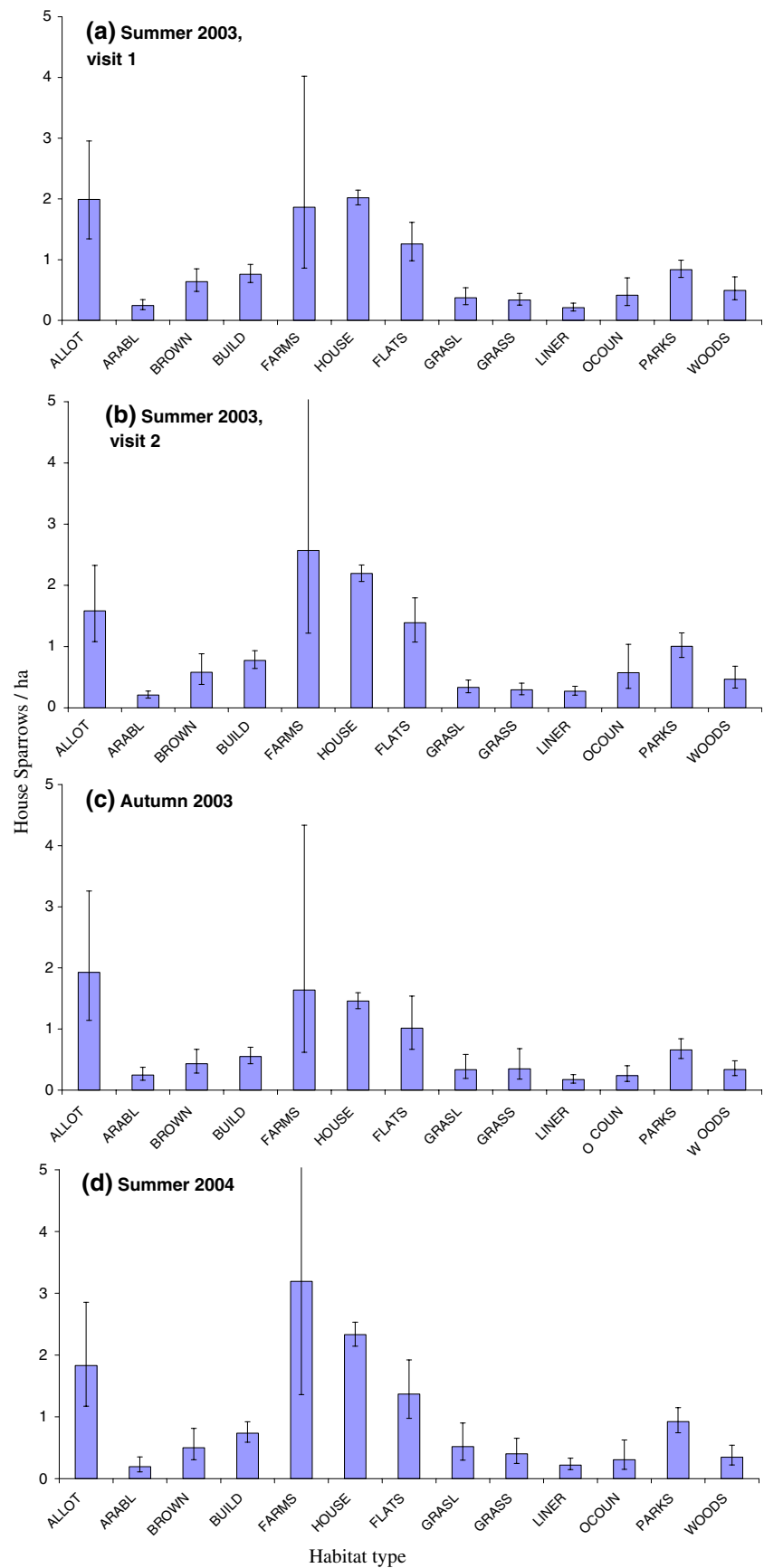


Table 3 Estimated house sparrow density (per ha) in the four habitats with the highest overall density (Figs. 1, 2)

	Visit 1	Visit 2	Visit 3	Visit 4
All house sparrows				
ALLOT	2.01 (1.35–2.99)	1.60 (1.09–2.33)	1.92 (1.15–3.20)	1.87 (1.20–2.92)
FARMS	1.84 (0.86–3.97)	2.57 (1.22–5.43)	1.64 (0.62–4.30)	3.17 (1.33–7.53)
HOUSE	2.01 (1.90–2.13)	2.19 (2.06–2.32)	1.45 (1.33–1.58)	2.32 (2.14–2.52)
FLAT	1.27 (1.00–1.63)	1.41 (1.09–1.82)	1.02 (0.68–1.54)	1.37 (0.98–1.91)
χ^2	234***	250***	48***	156***
Chirping male house sparrows				
ALLOT	0.24 (0.14–0.42)	0.20 (0.11–0.38)		0.16 (0.08–0.32)
FARMS	0.44 (0.12–1.62)	0.52 (0.19–1.41)		0.69 (0.23–2.05)
HOUSE	0.54 (0.50–0.59)	0.56 (0.52–0.61)		0.65 (0.60–0.72)
FLAT	0.43 (0.30–0.62)	0.33 (0.24–0.44)		0.47 (0.33–0.65)
χ^2	326***	342***		151***

Numbers in parenthesis are 95% confidence limits. Estimates were derived from a generalized linear model, fitted with negative binomial errors and including log(patch area) as an offset. The significance of the whole model is given by χ^2 with 4 *df* where *** $P < 0.001$. The significance of individual habitat types, derived from Z-tests (that compare untransformed estimates against 0), is indicated by estimates in bold text. No analyses were carried out for chirping male house sparrows in the autumn visit (Visit 3)

The total weight of each variable across all models (Σwi) is shown in Table 4, along with mean parameter estimates and 95% confidence limits. For all house sparrows, HOUSE was the single most important variable in terms of Σwi (Table 4, All house sparrows). This value was approximately 1, showing that any model without this variable contributed extremely little to the overall summed model weight ($wi < 10^{-5}$ in most cases). HOUSE is therefore clearly a key predictor of house sparrow abundance. There were four other variables that had high Σwi (> 0.95) and that, in common with HOUSE, showed confidence limits that did not overlap 0 indicating consistent effects across models. FLATS had almost as high a model weight as HOUSE. Other variables with consistent effects where $\Sigma wi > 0.95$ were ALLOT, BUILD and GRASL. These five variables (HOUSE, FLATS, ALLOT, BUILD and GRASL) are therefore the variables that are most closely associated with house sparrow abundance.

For chirping males, HOUSE and FLATS were the most important predictors over all models (Table 4). In contrast to the results for all house sparrows, chirping males were less consistently associated with ALLOT and GRASL, possibly suggesting that these are feeding rather than nesting habitats. Also, the variables FARMS and PARKS were relatively more important, suggesting the opposite (i.e. breeding rather than feeding habitats).

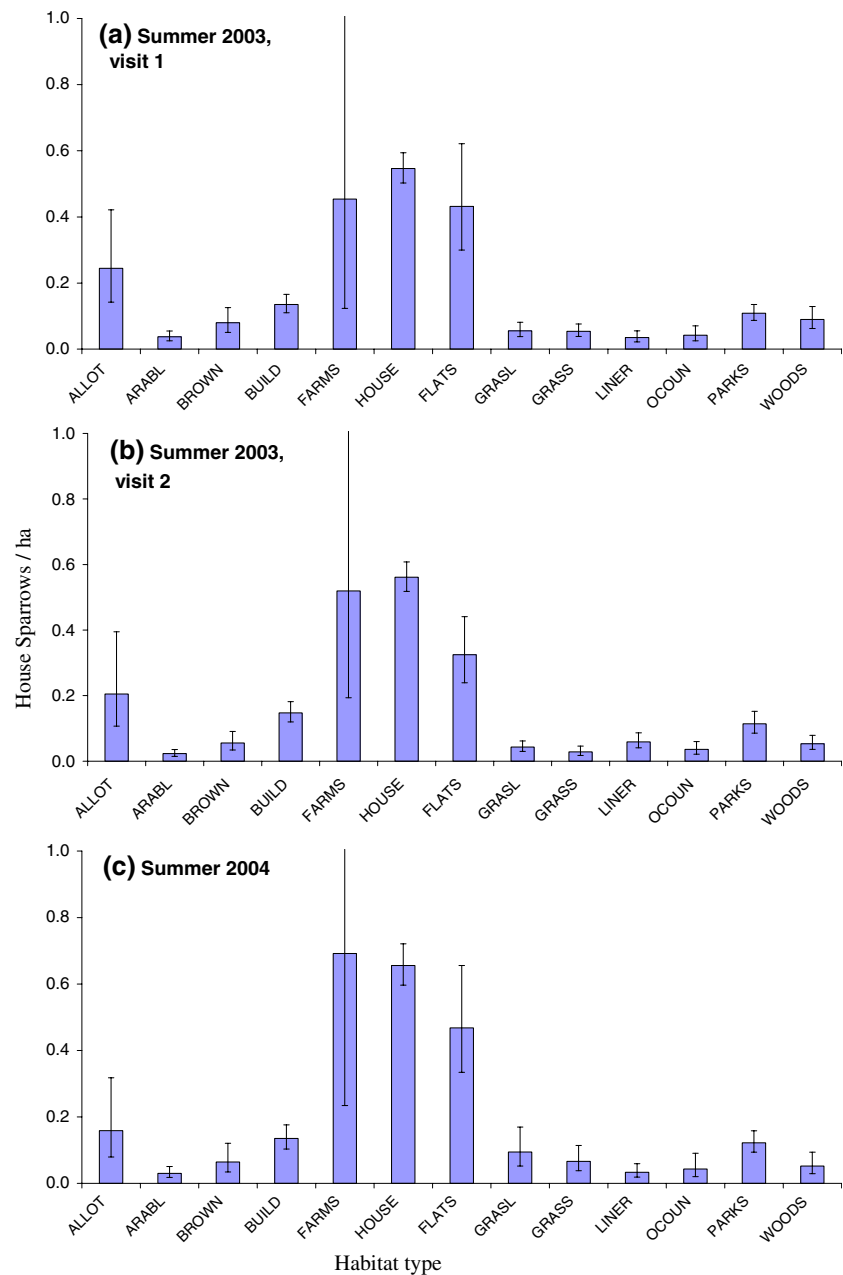
The modelling procedure was repeated separately for squares defined as high-density and low-density human cover according to the Landcover Map (LCM) 2000 data (Fuller et al. 2002). Variables showing consistent model effects (confidence limits not overlapping 0) for all house sparrows and chirping males in the two landscape types are

shown in Table 5. There were fewer variables with high Σwi , and estimates were generally less consistent compared to the model across all landscape types (Table 4), suggesting that some overall effects of certain variables (e.g. GRASS, PARKS, BUILD) may have been due to landscape-level differences in house sparrow abundance. Nonetheless, HOUSE had the highest Σwi , approaching 1 in all cases. There were, however, some landscape-specific differences. Allotments were a key predictor in high human cover landscapes for all house sparrows and chirping males. For chirping males, FLATS also showed consistent effects across models in both landscape types. In low human density landscapes, GRASL (all sparrows) and FARMS (chirping males) were also consistent predictors with relatively high Σwi .

Discussion

Residential areas, farm buildings and allotments were found to be key habitats for house sparrows. The high densities in areas of private housing and the consistent effect of the area of housing (Table 4) confirms earlier studies showing the importance of this habitat. Robinson et al. (2005) found that the highest densities of house sparrows occurred in suburban and rural housing, although there was a wide variation depending on region (1.5–5.2 birds/ha), and Heij (1985) found a positive correlation between house sparrow density and human population density across an urban–rural gradient. House sparrows occurred at higher densities when gardens were present within residential habitat. Both the area of residential

Fig. 2 Density of chirping male house sparrows in different habitat types in different years/seasons. *Error bars represent 95% confidence limits*



habitat with (HOUSE) and without (FLATS) gardens were key predictors of house sparrow abundance at the whole square level, but density increased with HOUSE far more steeply than with FLATS, with respective average model parameter estimates of 0.73 and 0.27 (Table 4). House sparrows are significantly more likely to occur where bushes are present (Wilkinson 2006), and gardens clearly form an important habitat for them. It was perhaps surprising that there was not a greater difference in house sparrow density between HOUSE and FLATS, but this may be due in part to the difficulties of surveying in residential areas. Some properties may have had gardens at the rear, but these may not have been detectable from public rights

of way. The relatively large errors on the density estimates for FLATS suggest a wide variation in the quality of this habitat type.

House sparrow density on allotments was one of the highest, and allotment cover was the second most important predictor at the whole square level for all house sparrows in high human cover landscapes, but it did not feature so prominently in low human cover landscapes. For chirping males, the density estimates were relatively lower, and allotment area was a predictor of relatively lower importance, suggesting that this habitat is more important for feeding than for breeding. Allotments may provide good foraging opportunities due to the diversity of the

Table 4 Results of multi-model inference for the influence of habitat extent of 13 habitat types on house sparrow density at the whole square level (500 × 500 m) for Visit 1

Habitat	Σwi	Parameter	LCL	UCL
All house sparrows				
HOUSE	1.000	0.725	0.610	0.840
FLATS	0.999	0.270	0.110	0.429
ALLOT	0.998	0.415	0.248	0.582
GRASL	0.990	0.167	0.054	0.280
BUILD	0.961	0.118	0.032	0.203
WOODS	0.925	-0.115	-0.201	-0.028
ARABL	0.903	0.099	0.011	0.186
PARKS	0.811	0.071	0.002	0.140
GRASS	0.765	0.068	-0.013	0.148
BROWN	0.594	0.062	-0.029	0.154
FARMS	0.512	0.154	-0.109	0.416
LINER	0.325	0.017	-0.029	0.063
OCOUN	0.275	0.0002	-0.032	0.032
Chirping males				
HOUSE	1.000	0.862	0.729	0.996
FLATS	1.000	0.386	0.217	0.556
BUILD	0.999	0.206	0.094	0.319
PARKS	0.993	0.165	0.073	0.257
FARMS	0.933	0.616	0.031	1.201
GRASS	0.932	0.130	0.025	0.235
ARABL	0.851	0.101	0.003	0.200
ALLOT	0.729	0.194	-0.011	0.399
GRASL	0.576	0.058	-0.038	0.153
OCOUN	0.558	0.053	-0.032	0.138
WOODS	0.396	-0.022	-0.077	0.033
BROWN	0.337	0.021	-0.039	0.081
LINER	0.337	-0.023	-0.089	0.043

Summed model weights (Σwi) give a measure of the importance of each variable over all models. Parameter estimates and 95% confidence limits (ULC, upper confidence limit; LCL, lower confidence limit) are derived from the mean estimates over all models. Habitat variables are given in order of Σwi . Estimates were derived from output for 4096 separate models (i.e. all combinations of 1 variable to 13 variable models)

habitat, not just in the sense of what is cultivated but also more generally, with abandoned or poorly maintained allotments likely to provide rich sources of both invertebrates and weed seeds.

The highest density estimates overall were for farm buildings (FARMS), and these were also key predictors at the whole square level in low human density landscapes. Rural housing has some of the highest estimated house sparrow densities (Robinson et al. 2005), and house sparrows have long been associated with farmyards. The increased cleanliness of farmyards and, in particular, the improved transport and storage of grain, coupled with a

Table 5 Results of multi-model inference for the influence of habitat extent of 13 habitat types on house sparrow density at the whole square level (500 × 500 m) for Visit 1 in high-density ($N = 392$) and low-density ($N = 555$) human cover landscapes. Only habitat variables where confidence limits don't overlap 0 are shown. Other details as per Table 4

Landscape	Habitat	Σwi	Parameter	LCL	UCL
All					
High density	HOUSE	0.999	0.438	0.216	0.659
	ALLOT	0.994	0.511	0.299	0.734
Low density	HOUSE	1.000	0.739	0.606	0.872
	GRASL	0.835	0.129	0.013	0.245
Chirping males					
High density	HOUSE	0.999	0.501	0.252	0.750
	ALLOT	0.916	0.438	0.174	0.703
	FLATS	0.759	0.187	0.005	0.370
Low density	HOUSE	1.000	0.950	0.790	1.110
	FLATS	0.965	0.670	0.256	1.084
	FARMS	0.942	0.796	0.041	1.550

lower availability of nest sites in modern buildings, could have been potential factors in the decline of farmland populations of house sparrow (Shrubb 2003). The high overall density recorded in this survey was also accompanied by a very large variation (Figs. 1, 2), suggesting a wide range in the quality of farm buildings for sparrows. Whether this is due to older farms versus more modern enterprises or whether there are other factors at work would seem a worthy topic for further research.

There was little evidence of strong seasonal patterns of habitat use when comparing visits. Counts were generally lower in the autumn, possibly due to both a higher detectability when breeding behaviour is shown and sparrows being more widely dispersed in the autumn. Despite the significant bias towards squares with higher house sparrow counts shown in the later visits, there was little evidence that this had much effect on habitat-specific density estimates. Counts were generally the highest in the final visit (Figs. 1, 2), but there was no apparent impact on the relative density estimates of different habitat types, suggesting that it is valid to use such data for habitat comparisons.

This study has demonstrated the importance of the suburban landscape for house sparrows: houses with gardens supported some of the highest densities, and the area of this habitat was the most consistent predictor of both breeding male house sparrows and of all house sparrows. This study is, to our knowledge, the first of its kind to demonstrate the importance of allotments to house sparrows within urbanized environments. Both of these habitats are likely to be under pressure due to increased demand for housing, specifically from the infilling of green space

within urban areas, such as gardens of large private houses and Victorian terraces sold off for building flats (Dawson and Gittings 1990), local authorities selling allotments and other amenity green space for property development.

An example of the potential impact of loss of garden habitat is shown in Fig. 3, where the predicted house sparrow count has been determined using model parameter estimates from Table 4 (All house sparrows), assuming that 10 ha of a 25-ha square is covered by HOUSE and FLATS combined and the remainder is a constant made up of the other habitat variables (the area of each being proportional to the relative mean area in Appendix A, Electronic Supplementary Material). As the ratio of FLATS:HOUSE increases, the predicted house sparrow count decreases rapidly, suggesting that even a relatively small loss of private gardens has large effects on sparrow abundance. Once the area of FLATS exceeds the area of HOUSE in the landscape, the decline slows, but by that point the house sparrow count has reached a low level. This model is an inevitable simplification and does not include potentially important effects such as habitat configuration (e.g. whether houses with private gardens are in homogeneous blocks or are dispersed through the landscape). Nevertheless, it illustrates that continued loss of private gardens within urban landscapes could have serious consequences for the house sparrow population. It would seem imperative that any action plan to protect urban house sparrow populations should include specific protection of such key habitats.

This study has provided some broad-scale but none-the-less essential information on habitat associations of house sparrows. The survey should act as an invaluable baseline against which to compare future population changes and to assess the impacts of habitat change. The results reported here also suggest where similar studies may be focussed in the future, specifically assessing the resource use by sparrows in private gardens and allotments and carrying out

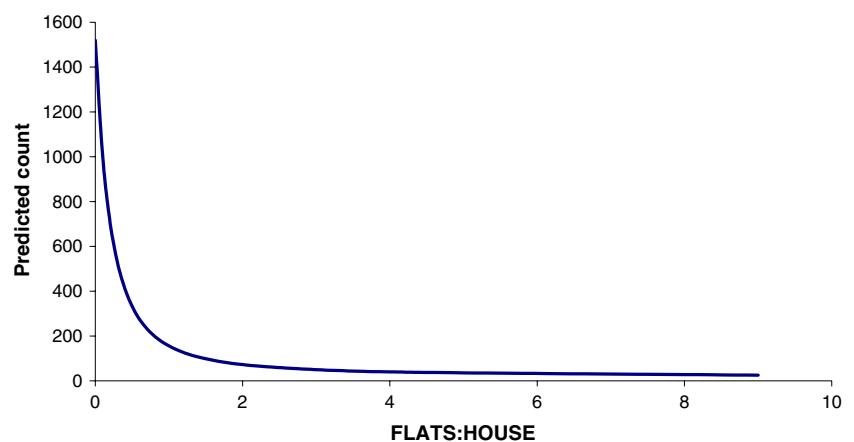
more intensive work to understand the wide variation in habitat quality of farm buildings.

Zusammenfassung

Habitatnutzung beim Haussperling (*Passer domesticus*) in urbanisierten Landschaften

In vielen Teilen Europas zeigt der Haussperling gravierende Bestandsrückgänge. Besonders stark sind diese in urbanen Gegenden, wobei bis jetzt relativ wenig bekannt ist über die Habitatbindung der Art in urbanisierten Landschaften. Wir untersuchten die Habitatbindung von Haussperlingen mit Hilfe einer Erhebung über 1,223 geschichtete und zufällig ausgewählte 500×500 m Quadrate innerhalb urbanisierter Landschaften, die mindestens 25% menschliche Nutzung aufwiesen. Die Dichte singender Männchen und aller Haussperlinge wurden einzeln analysiert, um Einsicht zu erhalten in die Anforderungen an Bruthabitate beziehungsweise an die generelle Habitatbindung. Multivariate Tests zeigten, dass Wohngebiete (Häuser, Appartements), Kleingärten und landwirtschaftliche Gebäude die Hauptfaktoren waren für die Vorhersage der Populationsdichte des Haussperlings und für die Dichte singender Männchen. Einzelne Analysen auf Flächen mit unterschiedlicher menschlicher Nutzung zeigten ähnliche Ergebnisse. Innerhalb von Wohngebieten war die Zunahme der Haussperlings-Dichte in Abhängigkeit von der bewohnten Fläche (auf einer logarithmischen Skala) ungefähr dreimal so groß, wenn private Gärten vorhanden waren als wenn sie nicht vorhanden waren. Dieses Modell sagte einen drastischen Rückgang in der Abundanz von Haussperlingen voraus, falls auch nur eine kleine Fläche von privaten Gärten in geschlossene Bebauung umgewandelt würde. Kleingärten und Wohngebiete mit privaten Gärten dürften zunehmend durch einen größeren Bedarf an Wohn-

Fig. 3 Predicted house sparrow count in a landscape where residential area with gardens (HOUSE) is replaced by residential area without gardens (FLATS). Predictions were based on model-averaged parameter estimates derived from Table 4 (All house sparrows), assuming a 25-ha square with a total area of HOUSE + FLATS of 10 ha (mean intercept = 1.40). Areas of other habitats were kept constant



fläche, insbesondere durch das Überbauen von Grünflächen in städtischen Bereichen, gefährdet sein. Der Erhalt dieser Schlüsselhabitate ist unbedingte Voraussetzung für den Schutz von Haussperlings-Populationen im städtischen Gegenden.

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